FM·RX

User manual

8-station FM band receiver · VST3 · version 1.0.0



NUWMUN

Windows 10/11 (x64) · macOS 10.13+ (Apple Silicon, Intel)

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1. What FM·RX is

FM·RX is a VST3 plug-in: a virtual FM band from 87.5 to 108 MHz. Up to eight stations broadcast on it at once — audio files or the audio of your track — and you tune between them like on a real radio: turn the dial, press SEARCH, catch a station through the noise.

Unlike typical "radio effects", the sound goes through **real frequency modulation**, the air and a receiver. Whistles and distortion when detuned, capture of the stronger station, a neighbour breaking through a wide filter and stereo collapsing on a weak signal all happen by themselves rather than being faked with filters.

What it's for

- "switching radio stations" intros and transitions in tracks and podcasts;
- an on-screen radio for film, games and sound design;
- lo-fi, hip-hop and ambient textures: air noise, fades, speaker colour;
- live transitions between song sections by automating the TUNE dial.

System requirements

Windows	Windows 10 or 11, 64-bit; a VST3 host
macOS	macOS 10.13 or later; Apple Silicon and Intel (universal binary); a VST3 host
Format	VST3 (effect, stereo in and out)
Sample rates	44.1–192 kHz
Latency	about 32 samples at 48 kHz (reported to the host automatically)
CPU	around 10–15 % of one core with three stations near the dial; distant stations cost almost nothing

2. Installation

Windows

1. Run `FM-RX-1.0.0-Windows-Setup.exe` and pick the installer language.
2. Confirm the administrator prompt — the plug-in goes to the shared folder for all users.
3. Click Install. On the last page you can open this manual right away.
4. Open your DAW and rescan plug-ins (see below).

Where the files go:

- plug-in — `C:\Program Files\Common Files\VST3\FM RX.vst3`;
- PDF manuals — `C:\Program Files\NUWMUN\FM RX`, with a shortcut in Start → NUWMUN.

Uninstall: Settings → Apps → FM RX → Uninstall. Update: just run the new installer over the old version.

macOS

1. Open `FM-RX-1.0.0-macOS.pkg`.
2. Follow the installer and enter your administrator password.
3. Open your DAW and rescan plug-ins.

Where the files go:

- plug-in — `/Library/Audio/Plug-Ins/VST3/FM RX.vst3`;
- PDF manuals — `/Library/Application Support/NUWMUN/FM RX`.

Uninstall: delete those two folders (in Finder: Go → Go to Folder…).

> If macOS says the installer is from an unidentified developer, right-click the file → Open, or go to System Settings → Privacy & Security → Open Anyway.

Rescanning plug-ins in your DAW

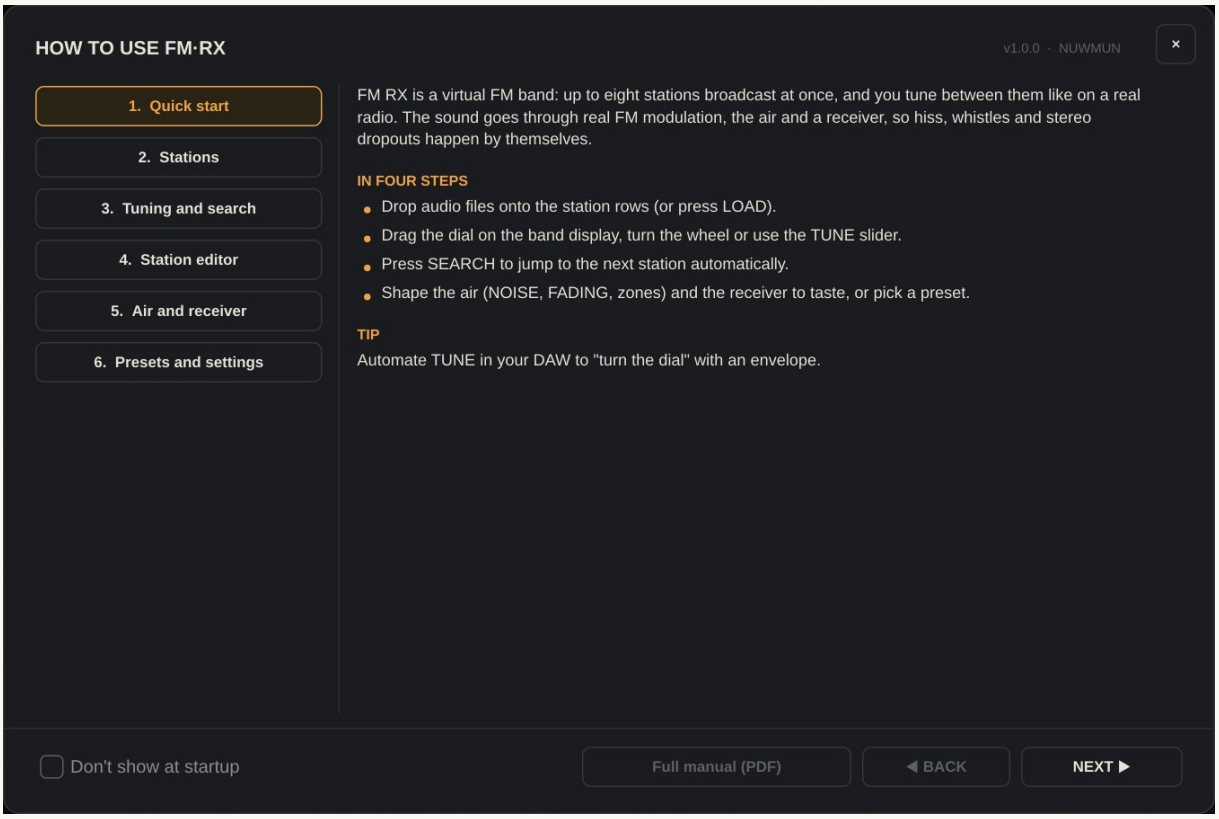
DAW	Where
Ableton Live	Settings → Plug-Ins: turn on "Use VST3 Plug-In System Folders" and press Rescan
FL Studio	Options → Manage plugins → Find installed plugins
Cubase / Nuendo	Studio → VST Plug-in Manager (usually found automatically on start)
Reaper	Options → Preferences → Plug-ins → VST → Re-scan
Studio One	scans on start; otherwise Options → Locations → VST Plug-ins
Bitwig Studio	Settings → Locations → Plug-ins: make sure the system VST3 folder is enabled

FM·RX shows up as an **effect** by **NUWMUN**. Insert it on any audio track or on an empty one — stations play regardless of the DAW transport.

3. First launch and help

The first time you open the plug-in window, a short help appears: six sections on the left, text on the right. Use NEXT / BACK or the arrow keys; Esc closes it.

- **Don't show at startup** — tick it and the help stops appearing by itself. Until it is ticked, the help opens the first time you open the plug-in in each DAW session (at most once per session).
- **Full manual (PDF)** — opens this document (installed by the installer).
- To bring the help back at any time: gear → **Help** → Open help.



The help window

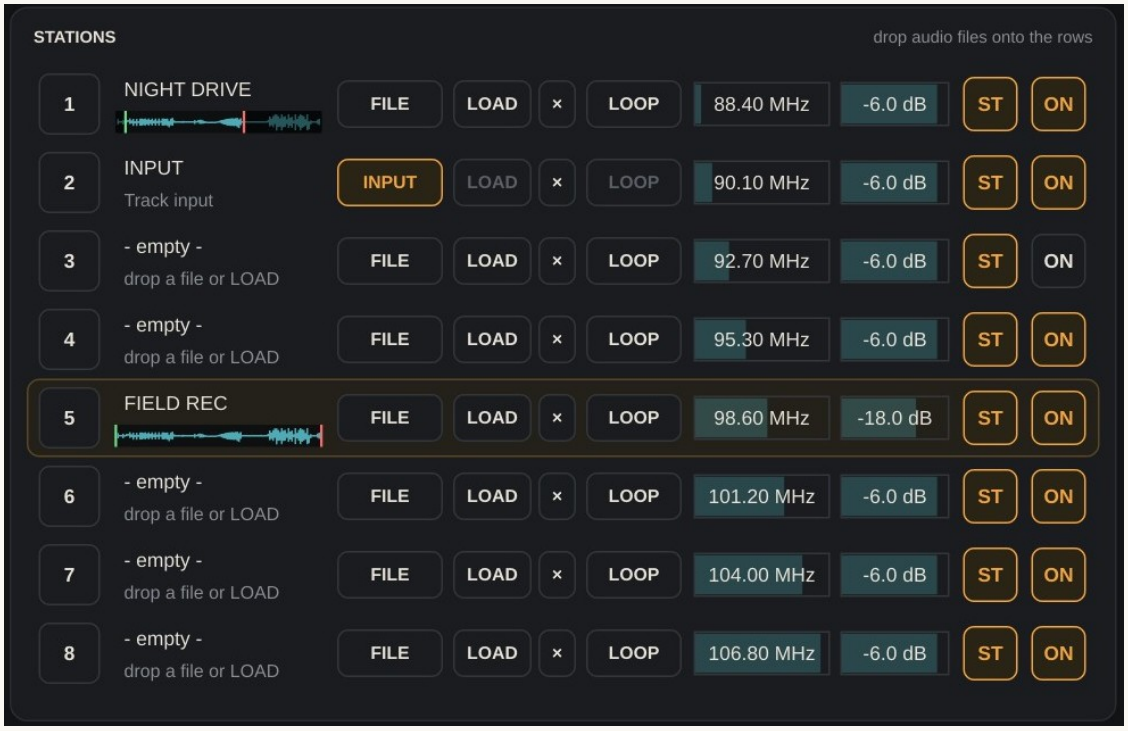
4. Window overview



1. **Presets** — step with the arrows; click the name for the list, saving and deleting.
2. **STEREO / MONO, AFC and the gear** — stereo mode, automatic frequency control, settings.
3. **Display** — frequency, station name (RDS), indicators, S-meter, SNR and AFC.
4. **Band display** — the stations, their reception zones and the tuning needle.
5. **Control strip** — SEARCH ◀ ▶, ±0.05 MHz steps, the TUNE slider, SCAN, STOP, LOCK, MUTE.
6. **SEARCH row** — search speed and sensitivity, button click volume.
7. **Stations** — eight transmitters; click a mini waveform to open the file editor.
8. **Air · channel** — noise, fading, multipath, interference, drift and reception zones.
9. **Receiver** — IF filter, stereo, soft mute, hi-cut, speaker, output, demodulator and de-emphasis.

Resize the window with the corner at the bottom right; the size is saved with the project.

5. Stations



The stations panel

Each of the eight rows is a separate transmitter with its own frequency and power. From left to right:

Control	What it does
Number	Tune to this station (the needle moves to its frequency).
Name	Double-click to rename (up to 16 characters). Shown on the display as RDS. Defaults to the file name.
Mini waveform	Overview of the file with the trim markers (green — start, red — end). Click it for the file editor.
FILE / INPUT	Source: an audio file, or the audio of the track the plug-in is on.
LOAD	Choose a file. You can also simply drop a file onto the row.
×	Remove the file from the station.
LOOP / 1-SHOT	Play in a circle or once (see section 7).
MHz	Station frequency, 87.5–108 MHz in 50 kHz steps. Drag; Shift — finer; double-click — type a value.
dB	Transmitter power, −60…0 dB. In FM a weak station is not quieter — it is noisier, less stable, and easier for a neighbour to override.
ST	Stereo broadcast (19 kHz pilot). Turn it off for a mono station.
ON	Station on air. A station that is off disappears from the band.

Files

- Formats: WAV, AIFF, MP3, FLAC, OGG. Mono or stereo.

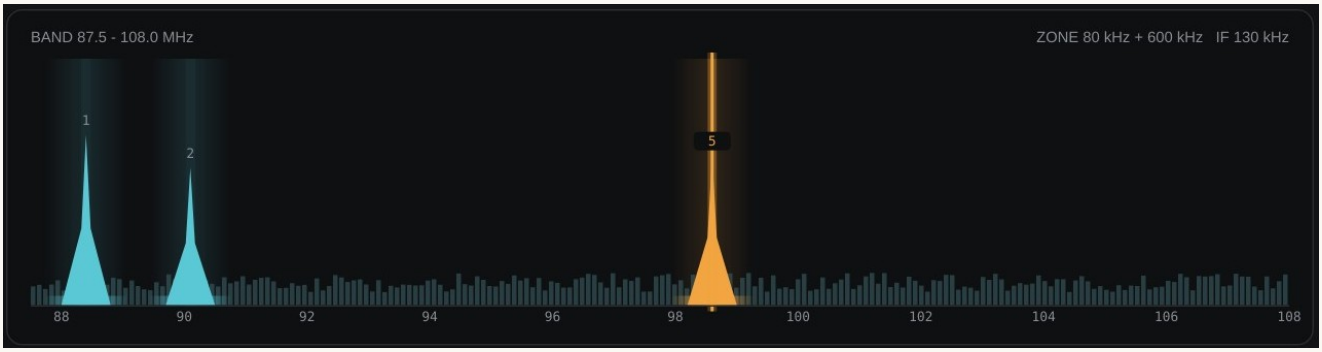
- On load a file is converted to the project sample rate and levelled (about −18 dBFS RMS), so stations sound even. Maximum length is 5 minutes; longer files are cut.
- Loading happens in the background without interrupting audio; the row shows "loading…" meanwhile.

> The project stores the **path** to the file, not the audio. If you move or delete the file, the row shows "file not found". Keep station files with your project, or use your DAW's collect-files feature before moving it.

Track input (INPUT)

In INPUT mode the station broadcasts the audio of the track FM·RX is on. Put the plug-in on a vocal, sampler or bus and switch one station to INPUT to put it "on air"; the other stations keep playing their files. LOOP/1-SHOT and the editor do not apply to an input station.

6. Tuning and search

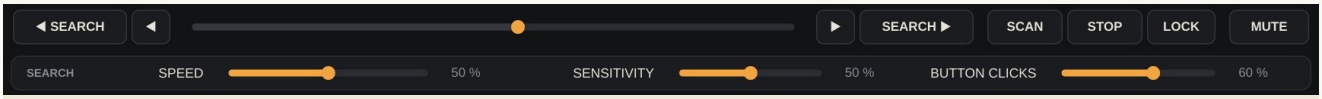


Band display: stations 1 and 2 in cyan, tuned station 5 in amber

The dial

- **Click a station peak** — tune exactly to it.
- **Click empty space** — the needle jumps there.
- **Drag** — a vernier: 1 pixel ≈ 10 kHz, five times finer with Shift.
- **Mouse wheel** — 50 kHz steps, 10 kHz with Shift.
- Under the peaks you see each station's **reception zone** (section 9); around the needle — the IF filter band.
- Marks next to the numbers: ■ — station stopped with STOP, ✓ — a 1-SHOT station has finished.

Control strip



Button	What it does
SEARCH ◀ / ▶	The dial moves down or up the band (wrapping around) until it meets a station the receiver can hear, then settles exactly on it. Weak stations are skipped. If a full circle finds nothing, the dial returns to where it started.
◀ ▶ (next to the slider)	±0.05 MHz steps.
TUNE slider	The tuning frequency. This is the **Tune** parameter — great for automation.
SCAN	Like search, but stops on every station for 5 seconds and moves on.
STOP	Stops search or scan where it is. The station you hear fades and rewinds to its start; it starts again when you tune away and back (or press ▶ FROM START in the editor).
Double STOP	Back to the very beginning: all stations from the start, needle to 87.5 MHz.
LOCK	Holds the frequency: the dial, search and automation cannot move the needle while LOCK is on.
MUTE	Silences the receiver output completely, air noise included. Switching it makes a double click.

Moving the dial during a search cancels the search. Button presses come with mechanical clicks.

SEARCH row

Slider	What it does
SPEED	How fast the dial moves while searching and scanning: 0.6 to 8 MHz per second.
SENSITIVITY	Which stations search stops on: higher catches weak ones too, lower — only solid reception.
BUTTON CLICKS	Volume of the mechanical clicks of SEARCH / SCAN / STOP / LOCK / MUTE. 0 — no clicks.

Automation

Every knob and switch is a plug-in parameter your DAW can automate. The most fun is **Tune**: draw an envelope and the needle sweeps the band along it, through noise and neighbouring stations. Search results are written back to Tune too, so the DAW always shows what you hear.

7. LOOP and 1-SHOT

LOOP

The file (or the region chosen in the editor) plays in a circle; the seam is smoothed with a short crossfade. The station broadcasts continuously even when you are not tuned to it — like real radio.

1-SHOT

1. The station waits at its start and begins when you tune to it (by hand or by search).
2. When it finishes, the station goes off air (✓ on the band).
3. Half a second later the receiver **starts a search by itself**, upwards from that frequency, and finds the next station.
4. If a full circle finds nothing above the threshold, the strongest available station is chosen. If there are no other stations, the same one plays again.
5. When you tune away from a finished station, it waits at its start again.

A few 1-SHOT stations make a "playlist with radio switching": each plays to the end and hands the air to the next.

> LOCK and SCAN keep the automatic search from moving the frequency. STOP cancels an automatic search in progress.

8. Station file editor



The editor opens in place of the station list

Click a station's mini waveform to open the editor. It trims the start and end of the file **without changing the file on disk**: the trim is saved with the project. Back to the list — ← STATIONS or Esc.

Control	What it does
START marker (green)	Where the sound starts. Drag the flag or the line.
END marker (red)	Where the sound ends. The darkened parts of the waveform do not play.
Times above the markers	Double-click to type an exact time, e.g. `0:02.500`.
Double-click a marker	Reset it to the start or end of the file.
Wheel / Shift+wheel	Zoom / scroll. The overview strip below shows the whole file and the visible part.
AUTO-TRIM SILENCE	Cuts silence at both ends using the SILENCE THRESHOLD (−80…−20 dB, default −50).
FADE-IN / FADE-OUT	Smooth fades at the edges of the region, 0–500 ms.
▶ FROM START	Tunes the receiver to this station and starts it from START — you hear it through the radio.
LOOP / 1-SHOT	The station's playback mode (same as in the row).
RESET	Whole file, default fades.
Name	Double-click to rename the station.

Markers snap to the nearest zero crossing, so the edges do not click. LOOP repeats exactly the START–END region, 1-SHOT plays it once, and STOP or re-tuning returns the station to START.

9. Reception zone

Two knobs in the Air · channel panel set how a station emerges from the noise as you tune towards it.

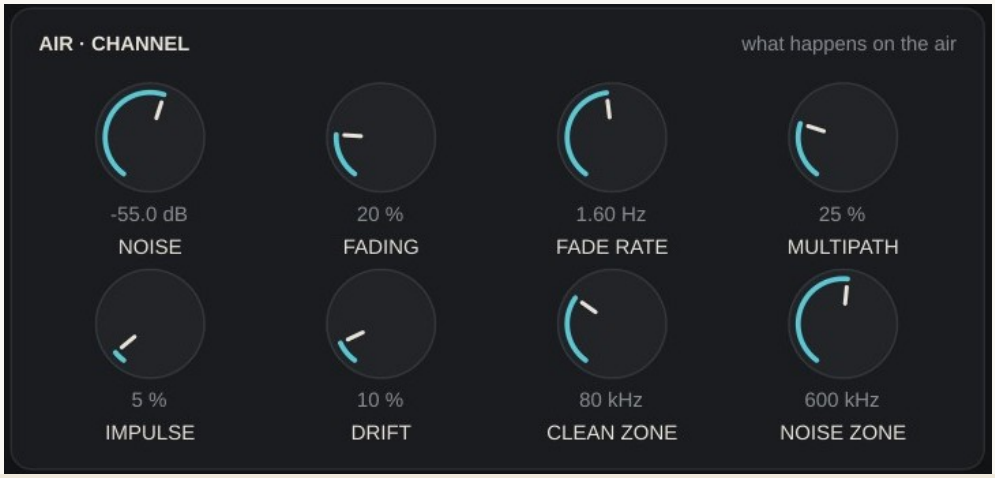
- **CLEAN ZONE** (0–1.5 MHz, default 60 kHz) — half-width of the area around a station where it arrives at full strength and sounds clean.
- **NOISE ZONE** (20 kHz–3 MHz, default 350 kHz) — width of the transition on each side: the station's level smoothly falls to −80 dB there and it sinks into the air noise.

The zones are drawn under each station on the band: the solid strip is the clean zone, the gradient is the noise zone. They follow the knobs.

- Narrow clean and narrow noise zones — stations "snap" in, like a digital tuner.
- Narrow clean and wide noise zones — a long emergence from the hiss, like slowly turning an old radio.
- Wide zones with SOFT MUTE off — long drifting transitions (the TUNING SWEEP preset).

Search uses the same zones: it stops where a station rises above the noise by more than the sensitivity threshold.

10. Air · channel



Knob	Range	What it does
NOISE	−100…−20 dB	Air noise level relative to the carrier. Higher — stations sink into the hiss sooner.
FADING	0–100 %	Depth of Rayleigh fading — the signal drops out and returns, as when moving.
FADE RATE	0.05–10 Hz	How often the fades happen.
MULTIPATH	0–100 %	A delayed reflection from buildings. Gives the typical distortion on fades.
IMPULSE	0–100 %	Impulse interference — crackles like a thunderstorm or a sparking motor.
DRIFT	0–100 %	Slow local oscillator drift: the station wanders. With AFC on, the receiver pulls it back.
CLEAN ZONE	0–1.5 MHz	See section 9.
NOISE ZONE	20 kHz–3 MHz	See section 9.

11. Receiver and display



Control	What it does
IF BW (40–200 kHz)	IF filter width. Narrower — fewer neighbours break through, but more distortion on loud parts. Wider — cleaner, but a neighbour may creep in.
BLEND	How early stereo falls back to mono on a weak signal.
SOFT MUTE	Hushes the hiss between stations and on weak signals.
HI-CUT	Cuts treble on weak signals — the noise gets softer.
SPEAKER	Small speaker colour: from full range to a pocket-radio sound.
OUTPUT (−24…+12 dB)	Output level. A soft limiter sits at the output.
QUAD / PLL	Demodulator type. They sound different when detuned and on weak signals — try both.
50 µs / 75 µs / Off	De-emphasis: 50 µs — Europe, 75 µs — USA, Off — no correction (brighter).
STEREO / MONO	Automatic stereo or forced mono (in the header).
AFC	Automatic frequency control: when the dial rests, the receiver pulls onto a strong station.

Display



- **STEREO** — lit while stereo is received.
- **TUNED / SEARCH / SCAN / LOCKED** — receiver state; blinks while searching.
- **MUTED** — lit while MUTE is on.
- Large digits — tuning frequency; below — station name (RDS).

- Bars — S-meter (signal strength); on the right — signal-to-noise ratio (SNR) and the AFC correction.

12. Presets

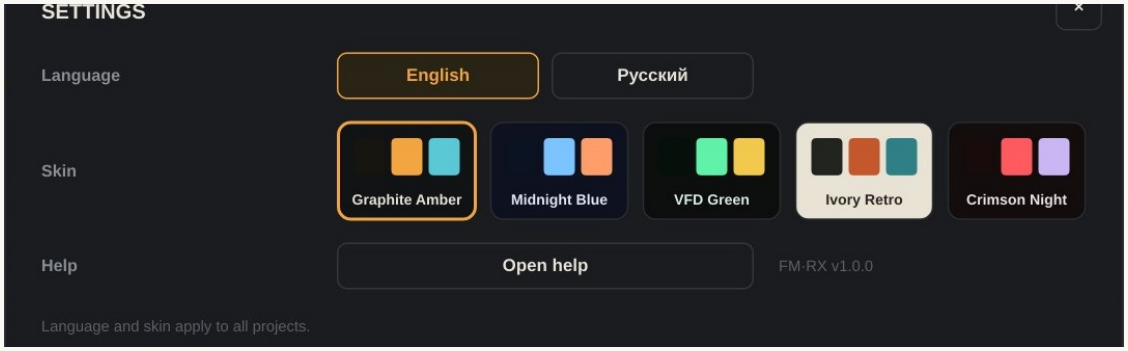
A preset sets the character of the air and the receiver: noise, fading, zones, filters, speaker, AFC. **Stations, frequency, MUTE and clicks are not changed by presets** — browse freely without losing your files.

- The ◀ ▶ arrows in the header step through presets; click the name for the full list.
- A star after the name (`NIGHT DRIVE *`) means you changed something after loading the preset.
- Save preset… stores your own; it appears under User in every project.
- Delete preset — for your own presets.

Preset	Character
CLEAN FM	Modern clean reception, almost no noise.
VINTAGE HI-FI	Warm 70s tuner: wide IF, gentle drift, no AFC.
CAR RADIO	On the move: fast fading, some multipath.
POCKET RADIO	Tiny mono speaker, narrow and boxy.
DISTANT STATION	Far away: the station barely rises above the hiss.
NIGHT DRIVE	Late highway: slow fades, soft instability.
HIP-HOP RADIO	Punchy radio intro: quick catches, grain, speaker colour.
LO-FI TAPE RADIO	Wobbly and dusty, like a radio recorded to cassette.
STORM	Thunderstorm: crackles, deep fades, unstable signal.
LOST SIGNAL	Signal drowning in noise; stations are hard to catch.
TUNING SWEEP	Wide zones and no mute: long noisy sweeps between stations.
DX OVERSEAS	Long-distance reception: narrow IF, slow deep fades.

User presets are `*.fmrxpreset` files: Windows — `%APPDATA%\NUWMUN\FM RX\Presets`, macOS — `~/Library/NUWMUN/FM RX/Presets`. You can copy them between computers.

13. Settings



The gear button at the top right

- **Language** — English or Russian. Switches instantly, no restart.
- **Skin** — five colour themes: Graphite Amber, Midnight Blue, VFD Green, Ivory Retro, Crimson Night.
- **Help** — open the help window again.

Language and skin apply to all projects and all instances. Everything else (knobs, stations, trims) is saved per project.

14. Tips and recipes

"Switching stations" intro

1. Load 3–4 short clips into stations, mode 1-SHOT.
2. Pick the HIP-HOP RADIO or CAR RADIO preset.
3. Tune to the first station — the receiver switches stations by itself as the clips end.

Transition between song sections

Put FM·RX on a bus, switch station 1 to INPUT and place another section or sample on a nearby frequency. Automate Tune from one station to the other — the transition passes through noise and whistles.

Radio "behind the wall"

SPEAKER 70–90 %, MONO, higher HI-CUT, NOISE around −45 dB — and your DAW's EQ after the plug-in.

Slowly hunting for a station

CLEAN ZONE 20–40 kHz, NOISE ZONE 600 kHz or more, SOFT MUTE at minimum, AFC off. Automate Tune with a smooth curve.

15. FAQ and troubleshooting

Problem	What to do
The plug-in is not in my DAW	Rescan plug-ins (section 2). Make sure the DAW is 64-bit and scans the system VST3 folder.
No sound	Check that stations have files and ON is lit; the needle is on a station (the display says TUNED); MUTE is off; OUTPUT is not at minimum.
Silence between stations instead of noise	That is SOFT MUTE. Turn it down if you want hiss between stations.
"file not found"	The file was moved or deleted. Press LOAD (or drop the file onto the row) and point to its new location — the trim and station name are kept.
A station wanders or distorts	Turn on AFC, lower DRIFT, widen IF BW, raise the station's power (dB).
Search skips a station	Raise SENSITIVITY or the station's power; lower NOISE.
Search does not stop	All stations may be too weak for the current NOISE — after a full circle the dial returns. Press STOP.
The help keeps appearing	Tick Don't show at startup in the help window.
High CPU	Load grows with the number of stations near the needle. Spread frequencies apart or turn off unused stations.

16. Automation parameters

These are the parameter names your DAW shows.

Parameter	Range	In the window
Tune	87.5–108 MHz	TUNE slider, band display
AFC	on / off	header
Stereo	Auto / Mono	header
Mute	on / off	MUTE
Noise	−100…−20 dB	Air
Fading	0–100 %	Air
Fade Rate	0.05–10 Hz	Air
Multipath	0–100 %	Air
Impulse	0–100 %	Air
Drift	0–100 %	Air
Clean Zone	0–1500 kHz	Air
Noise Zone	20–3000 kHz	Air
IF Bandwidth	40–200 kHz	Receiver
Demodulator	QUAD / PLL	Receiver
De-emphasis	50 us / 75 us / Off	Receiver
Stereo Blend	0–100 %	Receiver
Soft Mute	0–100 %	Receiver
Hi-Cut	0–100 %	Receiver
Speaker	0–100 %	Receiver
Output	−24…+12 dB	Receiver
Search Speed	0–100 %	SEARCH row
Search Sensitivity	0–100 %	SEARCH row
Button Clicks	0–100 %	SEARCH row
Station N On	on / off	station N: ON
Station N Frequency	87.5–108 MHz	station N: MHz
Station N TX Level	−60…0 dB	station N: dB
Station N Stereo	on / off	station N: ST
Station N Source	File / Input	station N: FILE / INPUT
Station N Playback	Loop / One-shot	station N: LOOP / 1-SHOT

17. About and licences

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The DSP algorithms (FM modulator, channel model, demodulators) are based on the open radiofm library © 2025 Henrik Forssell, MIT licence. The plug-in idea was inspired by the FM Radio Max for Live device by SSLL DSP.

Made with JUCE. VST is a trademark of Steinberg Media Technologies GmbH.